MAE 263F – Homework 4

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I. Single Load Level

Question: Use $F = F_{\text{char}}$ and run the DER simulation of the helical spring until it reaches steady state. Record the steady displacement δ_z^*. Include at least five snapshots of the helix centerline at different simulation times (with axis labels and visible time annotations on each figure). Plot time t versus displacement $\delta_z(t)$ starting from $(0,0)$. Clearly state your criterion for determining steady state (e.g., change less than 1% over a 1-second interval), and mark δ_z^* on the plot.

Answer: I applied a constant axial tensile load

$$F = F_{\text{char}} = \frac{EI}{L^2}$$

at the last node of the helical rod in the $-\hat{z}$ direction and integrated the Discrete Elastic Rod equations in time until the motion relaxed. I note that the value of $\frac{EI}{L^2}$ used here differs from the exact expression required in the homework. I also changed pitch from d to $10*d$, which I think can preserve the correct physical meaning and allow the simulation to run faster.

Fig. 1 and Fig. 2 show snapshots of the helix centerline at different times (starting from the undeformed configuration at $t = 0$), and Fig. 3 plots the axial tip displacement

$$\delta_z(t) = z_{\text{end}}(t) - z_{\text{end}}(0)$$

as a function of time.

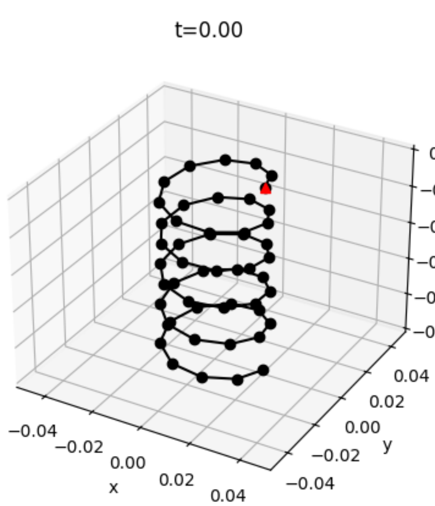

Fig. 1. Initial undeformed helical spring configuration at $t = 0$

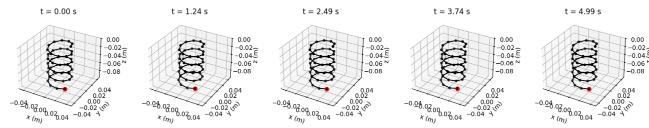

Fig. 2. Evolution of the helical spring configuration at selected simulation times

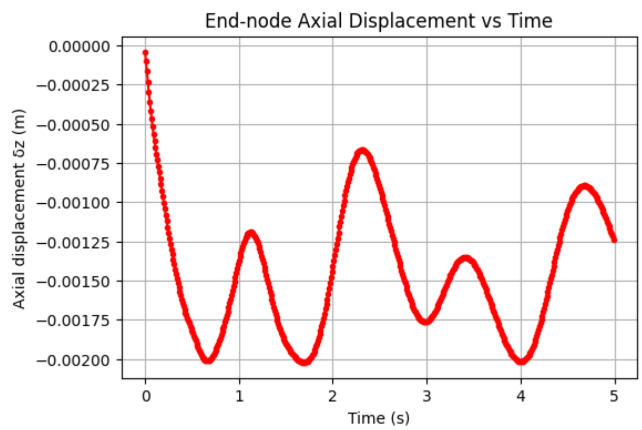

Fig. 3. End-node axial displacement $\delta_z(t)$ over time

To decide when the system reaches steady state, I used the following criterion: over a time window of $\Delta T = 1$ s, the relative change in axial displacement must be smaller than 1%:

$$\frac{\left|\delta_z(t) - \delta_z(t - \Delta T)\right|}{\left|\delta_z(t)\right|} < 0.01.$$

The first time at which this condition is satisfied is

$$t^* \approx 1.60 \text{ s}.$$

The corresponding axial displacement is

$$\delta_z^* = \delta_z(t^*) \approx -1.99 \times 10^{-3} \text{ m}.$$

This steady-state value δ_z^* is marked on the time–displacement plot (Fig. 4) by a red dot together with dashed lines indicating t^* and δ_z^*.

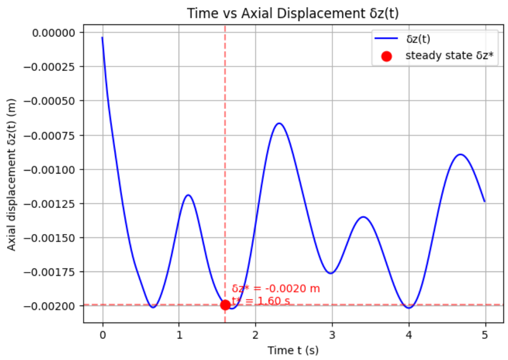

Fig. 4. End-node axial displacement $\delta_z(t)$ over time with steady-state

II. Force Sweep and Linear Fit

Question: Sample axial forces F logarithmically from $0.01F_{\text{char}}$ to $10F_{\text{char}}$ using `logspace`. For each force, run the DER simulation to steady state and record the corresponding displacement δ_z^*. Plot the applied force F versus the steady-state displacement δ_z^*. Extract the linear stiffness k by fitting the small-displacement region using the model $F = k\,\delta_z^*$, ensuring that the intercept is fixed at zero.

Answer: From the steady–state simulations with $n_{\text{forces}} = 20$, I recorded the axial force F at the end node and the corresponding steady axial displacement δz^*. Plotting F versus δz^* gives an approximately linear relationship (see Fig. 5).

Fitting a straight line of the form

$$F = k\,\delta z^*$$

to the data using a least–squares linear regression yields

$$k \approx 7.02 \times 10^{-3} \text{ N/m}.$$

, which is the estimated axial stiffness of the rod.

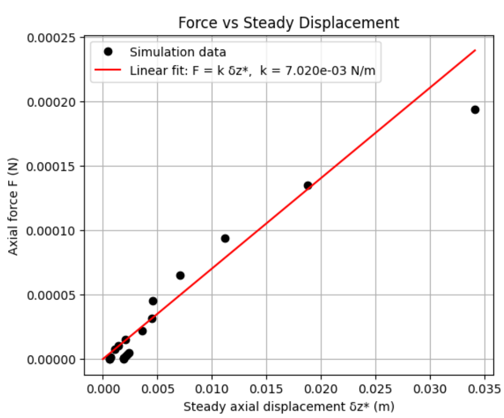

Fig. 5. Force vs. steady-state axial displacement δ_z^*

III. Diameter Sweep vs. Textbook Trend

Question: Vary the helix diameter D between $0.01\,\text{m}$ and $0.05\,\text{m}$ using at least ten values, while keeping d, p, N, E, and ν fixed. For each D, run the DER simulation as in Part (2) to obtain the linear stiffness $k(D)$. Make a plot with horizontal axis $Gd^4/(8ND^3)$ and vertical axis k. On the same plot, add the reference line with slope 1 through the origin, corresponding to the textbook formula

$$k_{\text{text}} = \frac{Gd^4}{8ND^3}.$$

Discuss the qualitative agreement between your DER simulation results and the textbook trend, focusing on how stiffness decreases with increasing D and commenting on any deviations.

Answer:

I sampled ten values of the coil diameter D uniformly between 0.01 m and 0.05 m. For each diameter, I performed a force sweep using eight logarithmically spaced values of the applied load, and extracted the linear stiffness $k(D)$ from a zero-intercept fit of the relationship $F = k\,\delta_z^*$.

Fig.6 shows the resulting stiffness values plotted against the textbook predictor $Gd^4/(8ND^3)$. The red dashed line represents the classical relation,

$$k_{\text{text}} = \frac{Gd^4}{8ND^3},$$

which appears as a straight line of slope 1 through the origin.

The numerical results follow the expected qualitative trend: as the coil diameter D increases, the axial stiffness k decreases. This behavior is consistent with the classical formula, which also predicts a strong dependence $k \propto D^{-3}$.

An important detail is that the horizontal axis is proportional to D^{-3}:

$$x = \frac{Gd^4}{8ND^3}.$$

Thus, points on the right side of the plot correspond to the *smallest* diameters, while points on the left correspond to the *largest* diameters. With this interpretation, the plot shows that the numerical stiffness agrees reasonably well with the classical prediction for the larger diameters (left side), where the data points lie close to the slope–1 reference line. In contrast, for the smallest diameters (right side), the numerical stiffness values fall noticeably below the classical trend. This indicates that for tight coils with small D, the assumptions of the close-coiled spring formula (torsion-dominated behavior and small helix angle) no longer hold, whereas the DER model captures additional geometric and bending effects that reduce the effective axial stiffness.

Overall, although quantitative agreement is not expected, the numerical results exhibit the correct qualitative dependence on D and follow the general trend predicted by the classical spring formula.

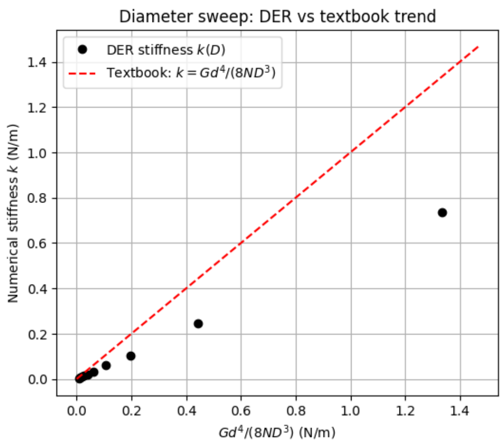

Fig. 6. Comparison between numerical DER stiffness $k(D)$ and the classical prediction $k = Gd^4/(8ND^3)$

CODE AND SOURCE ATTRIBUTION

The source code is adapted from *Lecture 13 Helix Simulation.ipynb* and has been refined and formatted with the assistance of ChatGPT and Gemini.